Project Design Phase – II

Technology Stack (Architecture & Stack)

Date	31 January 2025
Team ID	HYD-METRO-2025
Project Name	Hyderabad Metro Rail – Digital Ticketing System (ServiceNow)
Maximum Marks	4 Marks

Technical Architecture Overview

The Hyderabad Metro Rail Digital Ticketing System is built entirely on the ServiceNow platform (SaaS). The architecture follows a 3-tier model: Presentation (Service Portal), Application Logic (Scripts, Flows, Business Rules), and Data (Custom Tables, ACLs). No external servers or databases are required.

PRESENTATION LAYER (Frontend)	APPLICATION LOGIC LAYER (Middleware)	DATA LAYER (Backend)
Service Portal (/sp) Custom Metro Ticket Page Metro QR Ticket Widget Catalog UI (Book A Metro Ticket)	Catalog Client Scripts (JS) • Fare Auto-Populate (OnChange) • QR Code Generator (OnSubmit) • Field Validator (OnChange) Catalog UI Policies Flow Designer (Metro Project) Business Rules ACL Rules	Custom Table: Metro Station Details Custom Table: Metro Database ServiceNow System Tables: • sc_cat_item • sc_item_option • sc_req_item • sys_user

Table 1: Components & Technologies

S.No	Component	Description	Technology / Tool
1	User Interface	Commuter-facing web interface for ticket booking. Responsive portal accessible via browser on mobile or desktop.	ServiceNow Service Portal (AngularJS-based) Custom Portal Page: Metro Ticket Page Widget: Metro QR Ticket Widget
2	Service Catalog Item	Primary booking form with all variables for the commuter to fill in journey details.	ServiceNow Service Catalog (sc_cat_item) Item: Book A Metro Ticket
3	Application Logic – Client Scripts	JavaScript scripts running in the browser to auto-populate fares, generate QR codes, and validate fields.	ServiceNow Catalog Client Scripts Language: JavaScript (GlideForm API) Scripts: 3 (OnChange × 2, OnSubmit × 1)
4	Application Logic – UI Policies	Conditional display logic to show/hide form fields based on user's selection.	ServiceNow Catalog UI Policies Condition: Mode of Payment = 'Others' Action: Show 'Enter Payment Mode' field
5	Application Logic – Flow Designer	Process automation engine that captures catalog variables post-submission and creates records in the Metro Data	ServiceNow Flow Designer Flow: Metro Project Actions: Get Catalog Variables + Create Record

6	Application Logic – Business Rules	Server-side logic that triggers on Metro Database record insertion to auto-generate ticket IDs and send notifications	ServiceNow Business Rules Language: Server-side JavaScript (GlideRecord API) Rule: Generate Metro Ticket
7	Data Storage – Metro Station Details	Custom table storing all metro station master data referenced by the booking form.	ServiceNow Custom Table (cmdb_ci extension) Table Name: Metro Station's Details Fields: 6 (Name, Code, Line, Zone, Address, Active)
8	Data Storage – Metro Database	Custom table storing all booking transaction records auto-populated by Flow Designer.	ServiceNow Custom Table Table Name: Metro Database Fields: 10+ (Passenger, Source, Dest, Type, Fare, Payment, D
9	Access Control	Role-based security controlling who can read, write, create, or delete records in the Metro Database.	ServiceNow ACL Rules (sys_security_acl) Roles: admin, station_manager, itil Default ACLs auto-created with table
10	QR Code Generation	Client-side QR code generation triggered on form submission, displayed via GlideModal popup.	ServiceNow GlideModal API External QR API Page: external_qr_api_page QR URL: https://example.com/ticket?id=<sys_id>
11	Notification Engine	Automated email notifications sent to commuters upon successful booking.	ServiceNow Notification Engine Email Templates (HTML) Trigger: Metro Database record insert
12	Infrastructure	Platform is fully cloud-hosted as SaaS. No local server configuration required. All components are deployed on Ser	ServiceNow Cloud (SaaS) Instance Type: Personal Developer Instance (PDI) URL: https://<dev-instance>.service-now.com Hosting: ServiceNow multi-region cloud

Table 2: Application Characteristics

S.No	Characteristic	Description	Technology / Implementation
1	Open-Source Frameworks	ServiceNow Service Portal is built on AngularJS (open-source). GlideForm and GlideRecord APIs are Servic	• AngularJS (Service Portal base) • GlideForm API (client-side) • GlideRecord API (server-side) • GlideModal (popup/QR display)
2	Security Implementations	Security is enforced at multiple layers: Role-Based AC ... session manage	• Role-Based ACL Rules (sys_security_acl) • HTTPS / TLS encryption (platform-level) • ServiceNow session tokens • Client-side field validation (JS) • Server-side Business Rule validation
3	Scalable Architecture	The ServiceNow SaaS platform provides horizontal scalability automatically. Custom tables support millions	• ServiceNow SaaS auto-scaling • 3-tier architecture (Presentation / Logic / Data) • Stateless catalog client scripts • Flow Designer queue-based processing
4	Availability	ServiceNow provides 99.9%+ uptime SLA as a SaaS platform with multi-region data centers and automatic f	• ServiceNow SaaS 99.9% uptime SLA • Multi-region cloud deployment • Automatic failover (platform-managed) • No custom infrastructure to maintain
5	Performance	Performance is optimized by leveraging ServiceNow's built-in caching for portal pages and catalog items. Cl	• [illegible] • Asynchronous Flow Designer execution • ServiceNow portal page caching • GlideModal QR display ≤ 2 seconds • Flow completion ≤ 5 seconds

Technology Stack Summary

Category	Technology	Version / Notes
Platform	ServiceNow	Washington DC / Xanadu PDI
Frontend	ServiceNow Service Portal	AngularJS-based, responsive
Scripting	JavaScript (GlideForm, GlideRecord APIs)	Client-side & Server-side
Automation	ServiceNow Flow Designer	No-code/low-code process automation
Database	ServiceNow Custom Tables	Metro Station Details + Metro Database
Security	ServiceNow ACL Rules	Role-based (admin/manager/user)
QR Generation	GlideModal + External QR API	Client-side, on form submit
Notifications	ServiceNow Email Notifications	HTML email templates
Infrastructure	ServiceNow Cloud (SaaS)	Multi-region, 99.9%+ uptime
Deployment	ServiceNow Update Sets	Portable across instances